

The curious case of the two-headed phrase: The symmetrical syntax of co-compounds

Lena Borise (Université Paris Cité, CNRS, Laboratoire de linguistique formelle) & Tamás Halm (University of Vienna & Hungarian Research Centre for Linguistics)

Abstract

Co-compounds (CC) are pairs of morphosyntactically parallel and semantically related elements that constitute a single morphosyntactic and semantic unit. Usually described as a type of (asyndetic) coordination in the typological literature, they have only received sporadic attention from generative syntacticians, and have mostly been analysed as a subtype of exocentric compounds (Scalise, Fábregas & Forza 2009). In contrast to this previous analysis, based on data from two Uralic languages (Hungarian and Surgut Khanty), we argue that co-compounding is a syntactic operation: an instance of two heads undergoing Merge and being dominated by a shared layer of functional projections. While an implicit assumption of endocentricity (Chomsky 1970) and the projection principle (Chomsky 1993) has been that it is exactly one head that heads and projects a phrase, we argue that as long as a few conditions (that CCs are subject to) are met, the existence of such two-headed phrases is unproblematic for the standard understanding of these two core concepts. We also address the issue of labelling that emerges when two heads undergo Merge in a symmetric fashion. We model the semantics of CCs in terms of substitution by a shared superordinate term plus universal or existential closure. We also address variation between Hungarian and Surgut Khanty in terms of (i) the required degree of semantic relatedness between the two elements and (ii) the availability of dual/plural marking and a conjunctive interpretation; we relate this variation to the very different functional load that co-compounding carries in the two languages.

1. Introduction

A co-compound (CC) is a pair of morphosyntactically parallel and semantically related elements that constitute a single morphosyntactic and semantic unit. CCs (and CC-like structures) are also known under the names of *dvandva* compounds, pair words (Russian *parnye slova*), and copulative compounds (German *Kopulativkomposita*). CCs are well-described in the typological literature and are usually analysed as a type of (asyndetic) coordination (Wälchli 2005). They are amply attested in numerous languages, notably those spoken in Eurasia and Austronesia, including **Hungarian**¹ and **Khanty**², two languages belong to the Ugric branch of the Uralic language family. Some examples from both languages (boldfaced) are provided in (1) and (2):

- (1) a. *János **adta-vette** a használt autókat.* Hungarian
 John sold-bought the used cars
 'John was trading (lit. selling-buying) used cars.'
- b. *Anti **fel-alá**-sétált.*
 Tony up-down-walked
 'Tony was walking around (lit. up-down).'

¹ A terminological note is in order. In descriptive grammars of Hungarian, what we regard as CCs here have been divided into 3 categories :

(i) *álíkerszók* ‘fake echo words’: CCs the components of which are phonologically similar, such as *ázik-fázik* ‘get wet – get cold’;

(ii) *lazá szerkezetű mellérendelések* 'coordinations of a loose structure': CCs the components of which are phonologically dissimilar, such as *süt-főz* 'fry – boil';

(iii) *valódi mellérendelések* ‘real coordinations’ such as *rúgkapál* ‘kick – flail’: CCs the components of which are phonologically dissimilar, and only the last element receives inflectional suffixation.

For our current purposes, these distinctions are irrelevant. We analyze all CCs within categories (i-iii) simply as CCs.

² The Khanty data comes from elicitation with three native speakers of Surgut Khanty, unless otherwise noted; examples from textual sources are accompanied by references. For the sake of conciseness, and given that CCs are found in all Khanty varieties, we use the name ‘Khanty’ instead of ‘Surgut Khanty’ in the remainder of the paper.

- c. *János megosztotta velem ügyét-baját.*
 John shared me.with affair.3SG.ACC-problem.3SG.ACC
 ‘John shared all his goings-on (lit. affair-problem) with me.’
- d. *A volt barátok jobbra a maguk rohanós-sietős életével*
 the ex friends mostly the own running-hurrying life.with
vannak elfoglalta.
 are busy
 ‘Ex-friends are preoccupied with their own busy lives.’
- (2) a. ***je:ji-yən*** ***man’i-yən*** ***li:k-kən*** ***jin’i-yən.*** Khanty
 older_brother-DU younger_brother-DU eat-PST.3DU drink-PST.3DU
 ‘Older brother and younger brother ate and drank.’
- b. ***nuy-kən*** ***mi:n-yən***
 2SG-DU 1SG-DU
 ‘you and me’

CCs have only received sporadic attention from generative syntacticians, and have mostly been analysed as a subtype of exocentric compounds (Scalise, Fábregas & Forza 2009) – a classification which we would like to challenge in this paper. Building on Borise & É. Kiss (2022), we argue that co-compounding is a syntactic operation: an instance of two heads undergoing Merge and being dominated by a shared layer of functional projections.

This analysis has interesting potential theoretical implications, as an implicit assumption of endocentricity (Chomsky 1970) and the projection principle (Chomsky 1993) has been that it is *exactly* one head that heads and projects a phrase (cf. Lichte 2021 for a recent overview). We argue that as long as a few sensible conditions (that CCs are subject to) are met, the existence of a two-headed phrase is unproblematic for the standard understanding of endocentricity and the projection principle.

We also address the issue of labelling that emerges when two heads undergo Merge in a symmetric fashion. Such symmetric structures have been claimed to be avoided by natural languages because the resulting constituent cannot be labelled (e.g., Kayne 1994; Di Sciullo 2002; 2005): the labelling algorithm is looking for a lexical head, but in a symmetric structure minimal search is ambiguous, locating two heads. However, adopting an argument of Chomsky (2013: 43), we will propose that this problem does not arise if the two heads are (non-accidentally) identical in a relevant respect, providing the same label. As members of a CC necessarily share the same category and are also strictly identical featurally, the labelling problem does not arise.

Furthermore, we provide arguments against treating CCs as exocentric compounds and also against treating them as instances of asyndetic coordination. We will also contrast CCs to some superficially similar phenomena such as serial verb constructions (Baker 1989), coordinative bare singular nominals (Postma 1995), and low coordination (Abeillé 2006).

In terms of interpretation, we argue that CCs come in two types. So-called superordinating CCs are interpreted via substitution by a shared superordinate term, followed by universal or existential closure; on the other hand, so-called conjunctive CCs receive a conjunctive interpretation. Finally, we also address variation between Hungarian and Khanty in terms of (i) the required degree of semantic relatedness between the two elements and (ii) the availability of dual/plural marking and of a conjunctive interpretation. We relate this variation to the very different functional load that co-compounding carries in the two languages.

This paper is organized in the following way. In section 2, we introduce the similarities (2.1) and differences (2.2) between Hungarian and Khanty CCs. In sections 3 and 4, we provide arguments

against the two most commonly assumed analyses of CCs: CCs as exocentric compounds and CCs are instances of (asyndetic) coordination, respectively. In section 5 we lay out our analysis of CCs as consisting of two heads that undergo symmetric Merge and share higher functional projections. Additionally, in section 6, we provide evidence against other potential ways of approaching the analysis of CCs: namely, treating them as similar to serial verb constructions (SVCs), coordinative bare singular nominals (e.g., in Dutch), or instances of low/lexical coordination. Finally, in section 7, we outline some further research directions, and in section 8, provide a summary of our findings and conclude the paper.

2. Data

In this section, we introduce the empirical patterns displayed by CCs in Hungarian and Khanty, focusing first on the similarities and then also addressing the differences between the two languages.

2.1 Similarities between Hungarian and Khanty

A major hallmark of CCs is that they are made up of two juxtaposed lexical elements with no overt coordinator between them:

- (3) a. *ügy-é-t* (**és*) *baj-á-t* Hungarian
 affair-3SG-ACC and problem-3SG-ACC
 ‘his goings-on (lit. affair-problem)’
- b. *xə:nti xo* (**pə:nə*) *xə:nti ne:* Khanty
 Khanty man and Khanty woman
 ‘(Khanty) people (lit. Khanty man-Khanty woman)’

Notably, a CC can consist of at most two members; if there are more potential members, another phrase/clause is required:

- (4) *βoqu-yən tʃe:βər-yən təβ jot-ɐl jə-s-yən,* Khanty
 fox-DU rabbit-DU he with-3SG come-PST-3DU
kəmləy təβ jot-ɐl jə-s.
 wolverine he with-3SG come-PST.3SG
 ‘The fox [and] the rabbit came with him, the wolverine came with him.’ (The Ob-Ugric Database (OUDb))

Furthermore, the two members of a CC are obligatorily linearly adjacent³ and inseparable: when subject to movement (such as e.g. focus-induced movement of the verb), they move as a unit. This is illustrated for Hungarian in the examples in (5), which demonstrate the movement of a whole CC verb over the verbal particles *szept* and *meg*, respectively:

- (5) a. *János fúrta-faragta szept fúrta-faragta a szekrényt, nem pedig András.*
 John drilled-carved PRT the cupboard not although Andrew.

³ The only apparent exception to the strict adjacency requirement is the appearance of *se* ‘neither’ between the two elements, as in (i):

(i) *Jánosnak nem volt se kutya-se macskája.*
 John.DAT not was neither dog-3SG neither cat-3SG
 ‘John had no company whatsoever.’ (lit. ‘John had neither dog nor cat.’)

Under the reasonable assumption that *se* is a proclitic, we take its appearance in front of both elements of a CC to result from a PF operation: as far as syntax is concerned, there is only a single instance of [NEG] (just as there is only a single [3SG] head), however, at PF, the phonological exponents of [NEG] and [3SG] are copied on both elements. This is schematized in (ii):

(ii) *se kutya macska ja* ==> *se kutya-se macská-ja*
 [NEG] [canine company]-[feline company] [3SG]
 [NEG] [some company] [3SG] no company (of his) whatsoever

‘It was John who shredded the cupboard into pieces, not Andrew.’

- b. *Öt osztály saját maga sütötte-főzte meg ~~sütötte-főzte~~ ebédjét bográcsban*
 Five classes own self fried-boiled PRT lunch cauldron.in
vagy grillen.
 or grill.on
 ‘Five classes prepared their own lunch in a cauldron or as a barbecue.’

The two members of a CC are also closely related semantically: they are represented by synonyms (1d), taxonomic sisters (1c), antonyms (1b), or reverses (1a). Finally, strict morphological parallelism between the two members of a CC is required – e.g., inflectional morphemes or proclitics (see fn. 3) appear on both elements and must be fully matching:⁴

- (6) a. *ügy-é-t* - *baj-á-t* Hungarian
 affair-3SG-ACC problem-3SG-ACC
 ‘his goings-on (lit. affair-problem)’
 b. **ügy-e* - *baj-á-t*
 affair-3SG problem-3SG-ACC
 ‘his goings-on (lit. affair-problem)’
 c. **ügy-é-t* - *baj-a-í-t*
 affair-3SG-ACC problem-3SG-PL-ACC
 ‘(his goings-on (lit. affair-problems))’

Finally, in the presence of possessive marking, strict parallelism in possessor identity is also required (i.e., morphological parallelism in the absence of indexing parallelism is not allowed).

- (7) *i:mp-əł* *ke:jkə-yəł* Khanty
 dog-3SG cat-3SG
 ‘his/her_i dog & his/her_{i/*j} cat’⁵

2.2 Differences between Hungarian and Khanty

CCs in Hungarian and Khanty also exhibit a number of systematic differences: (i) the (un)availability of ‘contextual’ CCs, (ii) different patterns of number marking and (iii) the (un)availability of a conjunctive interpretation. Each of these differences is discussed in a dedicated subsection below.

2.2.1 ‘Contextual’ CCs

We have seen that, in general, the two elements of a co-compound are required to be closely related semantically (this is also the case in a number of languages beyond Hungarian and Khanty, cf. Wälchli 2005). Often, CCs consist of two elements that form a semantically stable, expected dyad: e.g., *woman-man*, *father-mother*, etc. However, Khanty also allows for what we call ‘contextual’ CCs, in which the two elements are not sisters in a semantic ontology; rather, they form a meaningful, coherent pair/dyad within the context of the actual discourse (e.g. the two main protagonists of a given story). It has to be noted, however, that even in the case of contextual CCs, a certain degree of semantic relatedness is required (8a), in the absence of which, a CC is not felicitous even in the presence of a unifying context (8b):

- (8) a. *ə:tji-yən* *tje:tji-yən* Khanty
 father-DU grandmother-DU
 ‘father & grandmother’
 (context: *Father and grandmother on father’s side are relatives.*)

⁵ Hungarian and Khanty both lack grammatical gender, which leads to gender ambiguity in examples like (7).

- b. ???*Kənjikə-yən sɔ:rt-yən pəsən aβtu-nə βot-yən.*
 book-DU pike-DU table top-LOC be-PST.3DU
 ‘A book and a pike were on the table.’

In contrast to Khanty, contextual co-compounds are strictly unavailable in Hungarian:

- (9) a. **Apá-d - nagyanyá-d hol van?* Hungarian
 father-2SG grandmother-2SG where is?
 intended: ‘Where is your father and your grandmother?’
 b. *Apá-d - anyá-d hol van?*
 father-2SG mother-2SG where is?
 ‘Where are your parents?’

2.2.2 Number marking

In Khanty, CCs that consist of countable nouns typically carry a dual (DU) suffix on each member (on the conjunctive interpretation, see below in 2.2.3). This, however, does not mean that each member of a CC refers to two items.⁶ Also, if used as a subject or topical object, a CC elicits dual agreement on the verb, as shown in (10):

- (10) *Mɛ: qut-əm-ɐ βoqi-yən ɐ:mp-yən jɔ:βət-yən.* Khanty
 1SG house-1SG-LAT fox-DU dog-DU come-PST.3DU
 ‘A fox and a dog came to my house.’

Dual marking may be omitted in the presence of other parallel morphology on both components of a CC (e.g., possessive suffixes), as shown by the examples in (11), both felicitous:

- (11) a. *i:mp-əm ke:ʃk-əm* Khanty
 dog-1SG cat-1SG
 ‘my dog and my cat’
 b. *ɐ:mp-ɣət-əm ke:ʃkɐ-ɣət-əm*⁷
 dog-DU-1SG cat-DU-1SG
 ‘my dog and my cat’

In contrast with countable nouns, mass nouns and nouns with generic meaning, if forming a CC, do not carry dual suffixes:

- (12) a. *ʎi:tot(*-yən) qu:t(*-yən)* Khanty
 food-DU fish-DU
 ‘food’
 b. *ru:t qantəy qoʎnə ɐ:rnəm βat-t?*
 Russian Khanty how further live-PRS.3SG
 ‘How will the Russians [and] the Khanty live on?’ (The Ob-Ugric Database (OUDb))

Interestingly, dual marking can also be found in non-nominal CCs:

- (13) *Togoś.ɣən tiis.ɣən ɣandiud.ən.na mola ɐndàs?* Khanty
 here.DU there.DU look.2SG.LOC what profit

⁶ To be more precise, the 2+2 interpretation of a CC that includes dual marking (so that the CC refers to the totality of four as opposed to two items) is possible and might be made more prominent in a particular context, but the 1+1 interpretation is the default one.

⁷ The dual suffix *-yən* is realized as *-ɣət-* in the context of the following possessive suffix.

‘What is the use of your looking here and there?’ (Wälchli 2005: 52)

Finally, plural instead of dual marking is used if each member of the CC refers to a plurality of objects:

- (14) *ju.γ-ət pam-ət* Khanty
 tree-PL grass-PL
 ‘trees and grasses’

In stark contrast with Khanty, in Hungarian, CCs do not allow for number marking:

- (15) a. *ügy-é-t* - *baj-á-t* Hungarian
 affair-3SG-ACC problem-3SG-ACC
 ‘his goings-on (lit. affair-problem)’
 b. **ügy-e-i-t* - *baj-a-i-t*
 affair-3SG-PL-ACC problem-3SG-PL-ACC
 (‘his goings-on (lit. affairs-problems)’)

2.2.3 Superordinating vs. conjunctive CCs

The third major difference between CCs in Hungarian and Khanty concerns semantic interpretation. In Hungarian, a CC as a whole refers to a single, superordinate concept and it has a quantificational interpretation (universal or existential), as in (15a). Building on (Bauer 2023) approach to CCs and the notion of hyponym-superordinate expressions, we call these CCs ‘superordinating’. In contrast, in Khanty, CCs can be either ‘superordinating’, as in (12), or ‘conjunctive’, as in (11), where the meaning of a CC is equivalent to that of coordination, and is close to the traditional meaning of *dvandva* (Bauer 2023). Only ‘conjunctive’ CCs in Khanty allow for number marking. We will return to these semantic differences in section 5.5.

3. Against CCs as exocentric compounds

Before advancing our own proposal, we would like to point out a number of shortcomings that other existent approaches to CCs face, in our estimation. The first approach to consider is the one that analyzes CCs as exocentric compounds.

An exocentric compound is a compound that is not a hyponym of either of its elements (e.g. a *pickpocket* refers neither to some sort of *pocket* nor to a type of *picking*, but rather, to a *person habitually engaged in picking other people’s pockets*). Scalise et al. (2009) developed a three-dimensional model of exocentricity. According to it, a compound is categorically exocentric if “the constituent in the head position does not impose its categorial features on the whole construction” (e.g. *killjoy* is a noun, even though *kill* is a verb). A compound is morphologically exocentric if “the morphological features of the compound are not identical to the morphological features of any of its constituents” (e.g. Italian *i [testa rasata]* ‘the skinheads’ is masculine plural even though both *testa* ‘head’ and *rasata* ‘shaven’ are feminine singular). Finally, a compound is semantically exocentric “if it denotes a class that cannot be derived from the classes denoted by its constituents” (in other words, if it violates any notion of semantic compositionality; e.g. in Spanish, the meaning of *la [media naranja]* ‘Mr/Mrs Right’ cannot be compositionally derived from the meaning of its components).

Pace Scalise et al. (2009), we argue that CCs cannot be taken to be exocentric compounds, as they are endocentric along all three dimensions proposed in their model. First, CCs are categorially endocentric: the two elements of any CC necessarily have the same categorial features and the CC as a whole inherits this identical set of categorial features from its two elements. As long as one is willing to entertain the possibility of a compound having two heads, this means compliance with categorial

endocentricity. CCs are also morphologically endocentric: as we have seen above, the morphological features of the whole construction are strictly identical to the morphological features of its internal constituents.⁸ Finally, CCs are also semantically endocentric: whether superordinating or conjunctive, their semantic type and their meaning can be compositionally derived from the type and meaning of their constituents. In sum, this means that approaching CCs as exocentric compounds cannot be maintained.

4. Against CCs as (asyndetic) coordination

In this section, we provide evidence against approaching CCs as instances of asyndetic coordination. As we have seen above, overt coordinators are strictly prohibited in CCs, (3). In addition to this, we show below that postulating a silent coordinator between elements of a CC would both predict unavailable interpretations and exclude available interpretations. The pattern of dual marking in Khanty will also be shown to be incompatible with coordination. We point out, too, that CC-formation differs starkly from coordination in requiring strict morphological parallelism between the components of a CC, as well as possessor identity, if possessive marking is present. Finally, we draw attention to the fact that CCs necessarily share their complements and modifiers: accounting for this pattern would be unduly complicated in any analysis in terms of coordination (e.g., by postulating ellipsis).

4.1 Interpretation without coordination

In superordinating CCs, the meaning of the CC is a single concept. As we will argue below, this meaning is derived not via coordination (disjunction or conjunction), but rather, through substitution by an immediate shared superordinate term. First, consider a Hungarian example in (16):

- (16) *János a konyhában süit-főz.* Hungarian
 John the kitchen.in fries-cooks.
 ‘John is preparing dishes/cooking in the kitchen.’

Crucially, (16) can be uttered truthfully if John is in the kitchen preparing some dish (e.g., a cold salad) without either frying it (boiling it in oil/fat) or cooking it (boiling it in water). This means that LF₁ and LF₂ in (17), which correspond to the conjunctive and disjunctive types of coordination, respectively, are not the correct representations of the meaning of (16). Only LF₃ in (17) produces the correct interpretation:

- (17) PF: *János a konyhában süit-főz.* Hungarian
 John the kitchen.in fries-cooks.
 ✗ LF₁: FRY(**John**) ∧ COOK(**John**)
 ‘John is frying and cooking some dishes in the kitchen.’
 ✗ LF₂: FRY(**John**) ∨ COOK(**John**)
 ‘John is frying or cooking some dishes in the kitchen.’
 ✓ LF₃: DISH-PREPARE(**John**)
 ‘John is preparing dishes in the kitchen.’

⁸ Scalise et al. (2009) consider (nominal) CCs morphologically exocentric because, in many languages, the gender of the CC is neuter when the constituents have non-identical gender. Since Uralic languages have no grammatical gender, this switch is not attested in Hungarian or Khanty; accordingly, there is no reason to assume that CCs are morphologically exocentric in these languages. More generally, in languages where such a switch does occur, it appears to be rule-based: switch to neuter iff the gender of the constituents is non-identical – arguably, this could be handled as a resolution mechanism of conflicting feature values and does not necessitate the stipulation of an external morphological head.

Similar evidence is available in Khanty. As shown in (18), the concept of mankind/humanity/people in general in Khanty can be expressed as *rut'-qāntəy* 'Russian-Khanty'. However, as this example also shows, *rut'-qāntəy*, given its meaning as a superordinate CC, also includes other nations:

- (18) (Context: Khanty are at war with Selkups. The Khanty protagonist is surrounded by Selkup warriors and thinks to himself):
- | | | | | | | | |
|--------------|----------------|------------------|-----------------|-----------|---------------|-------------|---------------|
| <i>nām-a</i> | <i>rut'-nə</i> | <i>qāntəy-nə</i> | <i>wāl-tayə</i> | <i>ja</i> | <i>kič-əm</i> | <i>əntə</i> | <i>wǎa-a.</i> |
| in.vain | Russian-LOC | Khanty-LOC | kill-INF | PCL | desire-1SG | NEG | be-PRS.3SG |
- 'It's of no use, being killed by other people.' (Lit.: 'My desire to be killed by other people in vain doesn't exist.') (Márta Csepregi, personal archive)

It is clear that *rut'-qāntəy* is not an example of coordination, either on a conjunctive or disjunctive reading, since, in (18), the most salient representatives of mankind that pose a threat to the protagonist are Selkups (i.e., neither Russians nor Khanty). The semantic content of the CC is schematized in (19):

- (19) ✗ LF₁: RUSSIANS $\wedge$ KHANTY
 'It's of no use, being killed by Russians and Khanty.'
 ✗ LF₂: RUSSIANS $\vee$ KHANTY
 'It's of no use, being killed by Russians or Khanty.'
 ✓ LF₃: MANKIND/OTHER PEOPLE
 'It's of no use, being killed by other people.'

As these examples demonstrate, the interpretation of (superordinating) CCs does not involve either conjunction or disjunction at LF, which makes the assumption of coordination at the level of morphosyntax highly implausible.

4.2 The interpretation of dual marking in Khanty

Outside the context of CCs, Khanty marks cardinality in a singular-dual-plural system. By default, dual number in Khanty marks a cardinality of two, as shown in (20a). Accordingly, in the context of coordination, dual-marked nominals refer to two individuals/items each, as shown in (20b).

- (20) a. *Ma qut-əm-v (kə:t) βoqi-yən jv:βət-yən.* Khanty
 1SG house-1SG-LAT two fox-DU come-PST.3DU
 'Two foxes came to my house.'
- b. *Ma qut-əm-v (kə:t) βoqi-yən pə:nə (kə:t) v:mp-yən jv:βət-yən.*
 1SG house-1SG-LAT two fox-DU and two dog-DU come-PST.3DU
 'Two foxes and two dogs came to my house.'

Dual marking in CCs, in contrast, does not function as ordinary number marking: here, each of the dual-marked members of a CC typically refers to one item, as shown in (21), repeated from (10):

- (21) *Me: qut-əm-v βoqi-yən v:mp-yən jv:βət-yən.* Khanty
 1SG house-1SG-LAT fox-DU dog-DU come-PST.3DU
 'A fox and a dog came to my house.'

This interpretation would be unexpected if CCs were examples of (asyndetic) coordination.

4.3 Morphological parallelism and possessor identity

The strict morphological parallelism between the members of a CC, as discussed in section 2.1, is also not straightforward to explain on a coordination analysis. In the case of (overt) coordination, no such

parallelism is required in Khanty, as shown in (22b); if the same two nominals were to form a CC, however, that would result in infelicity, as shown in (22a).

- (22) a. **i:mp-əm ke:/kə-γəl-əm* Khanty
 dog-1SG cat-DU-1SG
 (intended: ‘my dog and my two cats’)
- b. *i:mp-əm pə:nə ke:/kə-γəl-əm*
 dog-1SG and cat-DU-1SG
 ‘my dog and my two cats’

Similarly, the requirement for possessor identity, already demonstrated in (7) in section 2.1, only holds in CCs but not in (overt) coordination in Khanty:

- (23) a. *i:mp-əl ke:/kə-γəl⁹* Khanty
 dog-3SG cat-3SG
 ‘his/her_i dog and his/her_{i/*j} cat’
- b. *i:mp-əl pə:nə ke:/kə-γəl*
 dog-3SG and cat-3SG
 ‘his/her_i dog & his/her_{i/j} cat’

These patterns also cast serious doubt on any coordination analysis.

4.3 Shared complements & functional projections

As a final piece of evidence against a coordination analysis of CCs, we show that the components of a CC necessarily share their complements and modifiers, which would be unexpected on a coordination analysis. To start with, in Hungarian, verbal particles (and verb modifiers in general) are standardly analysed as phrasal complements to the verbal head (Piñón 1995; É. Kiss 2002; Den Dikken 2004, a.o.). A verbal CC selects for a single verbal particle or verb modifier, which shows that the elements of a verbal CC in Hungarian cannot have independent complements:

- (24) *János el-tett - (*el-)vett a konyhában.* Hungarian
 John PRT-put.down PRT-pick.up the kitchen.in
 ‘John whiled away the time by moving (lit. putting down - picking up) stuff around in the kitchen.’
- (25) *A bizottság tagjai meg-hányták - (*meg-)vetették a kérdést.*
 the committee members PRT-threw PRT-cast the problem
 ‘The members of the committee thrashed out (lit. threw-cast) the problem.’
- (26) *A szocialista realizmus egyik legkifejezőbb alkotását*
 the socialist realism one most.expressive artefact
*a tömeg darabokra törté - (*darabokra) zúzta.*
 the crowd pieces.into broke pieces-into crushed
 ‘The crowd smashed into pieces one of the most expressive artefacts of socialist realism.’

Furthermore, a nominal CC can only associate with a single possessor, (27), and an adjectival CC can only be modified by a single adverbial, (28):

- (27) a. *Összegyűlt a falu fiatal-ja-öreg-je.* Hungarian
 assembled the village young-3SG-old-3SG
 ‘The whole population (lit. the young-the old) of the village assembled.’

⁹ /γ/ in the possessive suffix in *ke:/kə-γəl* is used for vowel hiatus resolution.

- b. *Összegyűlt a falu fiatal-ja - a város öreg-je.
 assembled the village young-3SG the town old-3SG
 (intended: ‘The young people from the village and the old people from the town assembled.’)

- (28) *Mari kicsit bús - (*kicsit) komor volt.* Hungarian
 Mary slightly gloomy slightly sorrowful was.
 ‘Mary was a bit sad.’ (lit. ‘Mary was a bit gloomy-sorrowful.’)

Similarly, a modifying adjective necessarily applies to both elements of a CC, as illustrated for Khanty in (29):

- (29) *M̨e: ʌnəl sp:rt-ɣən v:ɣər-ɣən qv:tl-əm.* Khanty
 1SG big pike-DU ide-DU catch-PST.1SG
 ‘I caught a big pike and [a big] ide.’
 (NOT: ‘I caught a big pike and an ide [of unspecified size].’)

Combined, (24)-(29) show that the members of a CC share their complements as well as modifiers and higher functional projections, which would complicate the coordination analysis, but falls out straightforwardly from the analysis that we will advance below.

5. Analysis

5.1 Merge & labelling

We propose that the derivation of a CC takes place in the syntax and proceeds as follows. The two elements of the CC enter the numeration as separate elements, which are then combined via symmetric Head-Head Merge ($[_\alpha H_1 H_2]$). The two heads are thus equidistant from α , the immediately dominating node, but since they are of the same category, the unresolved competition for which head should provide the label for the next level of structure is not problematic: H_1 and H_2 trivially contribute the same category to α . If H_1 and H_2 have identical subcategorization properties and theta-grids, they together project the (extended) XP. Otherwise, the derivation crashes. Further in the derivation, α functions as a single head for the purposes of e.g. movement. For the purposes of post-syntactic suffixation, which we take to proceed via M-Merger (more on this below), both heads are visible.

5.2 In favour of symmetry

We take the members of a CC to form a symmetric structure, as in (30b); this is similar in spirit to, e.g., (Fábregas 2020) approach to Spanish *dvandvas*. In contrast, coordinated expressions are typically assumed to be asymmetric (Munn 1987; Johannessen 1996), as in (30a):

- (30) a. regular coordination b. co-compounding
- ```

 &P
 / \
 VP &'
 / \
 & VP

```

```

 V
 / \
 V V

```

Despite the asymmetric analysis being prevalent for coordinated structures, as we show below, neither the empirical evidence of asymmetry nor theoretical arguments in its favour are valid for CCs. On the empirical side, the lack of overt conjunctions means that there is no prosodic evidence for an asymmetric c-command relation between the members of the CC, which could be signalled by the formation of a closer prosodic unit between the conjunction and one of the CC-members.

On the theoretical side, symmetric structures are claimed to be avoided by natural languages because the resulting constituent cannot be labelled (e.g., Kayne 1994; Di Sciullo 2002; 2005): the labelling algorithm is looking for a lexical head, but, in a symmetric structure, the minimal search is ambiguous, locating two heads. However, as Chomsky (2013: 43) argues, this problem does not arise if the two heads are (non-accidentally) identical in a relevant respect, providing the same label. Members of a CC necessarily share the same category and are also strictly identical featurally, so that the labelling problem does not arise.<sup>10</sup>

### 5.3 Linearization

A further problem associated with symmetric structures is that they cannot be linearized. To address it, we propose that the linearization of a CC is indeed free in the syntax; it is determined either semantically, based on importance/primacy — e.g., *eat-drink* in (2a) — or phonologically. Specifically, Hungarian CCs display a uniform phonological pattern: if one of the members is a front-vowel word and the other one is a back-vowel one<sup>11</sup>, then the front-vowel member precedes the back-vowel one:

- (31) *üt - vág* Hungarian  
 hit chop  
 ‘to beat indiscriminately’ (lit. ‘hit-chop’)
- (32) *tejbe - vajba (fürdet)*  
 milk.in butter.in bathe  
 ‘to provide for luxuriously’ (lit. ‘to bathe in milk – in butter’)

Otherwise, the shorter member precedes the longer one (Sóskuthy 2012; Patay 2018):

- (33) *boldog - boldogtalan*  
 happyhappy.less  
 ‘everybody’ (lit. happy-unhappy)

Importantly, however, that this pattern is relatively easy to override (e.g., in poetry, for reasons of rhyme or metre):

- (34) *Fürödtünk vajba-tejbe, de én nem akartam, hogy én legyek az /*  
 bathed butter.in-milk.in but I not wanted that I be that  
*Akit a kalapácsos tegnap este először vág fejbe.*  
 whom the person.with.hammer yesterday evening first hit head.in  
 ‘We were having the time of our lives (bathing in butter-milk) but I did not want to be the  
 one whom the guy with the hammer would hit in the head first yesterday evening.’  
 (lyrics of the song *Ki csinál rendet?* [Who shall create order?] by the punk rock band  
 Tankcsapda)

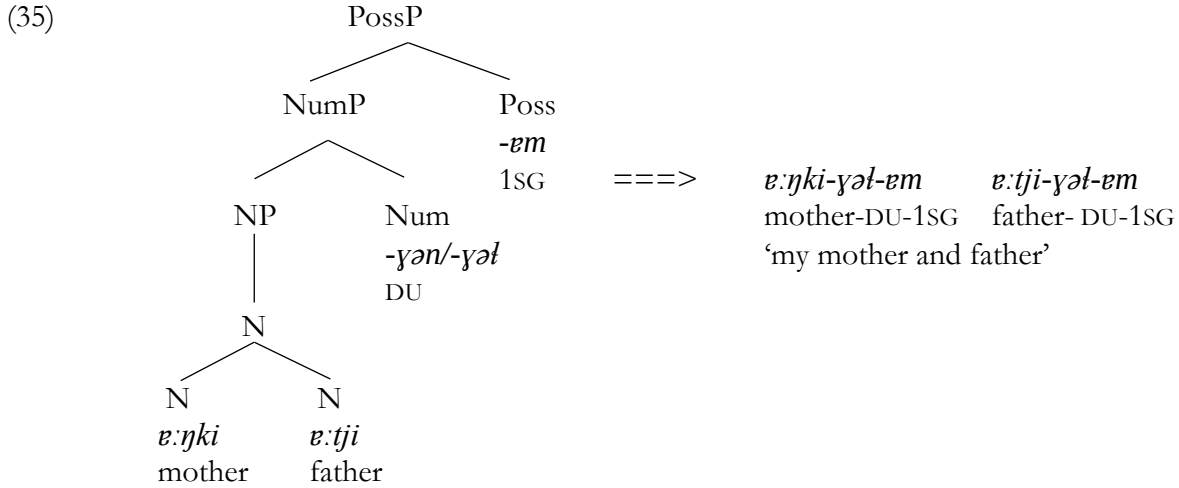
This supports the idea of a syntactically free, semantically and/or phonologically conditioned linearization.

<sup>10</sup> Chomsky (2013) posits the requirement of *non-accidental* identity in order to explain why two DPs merged in an XP-YP structure (e.g. in a small clause with a predicate nominal) cannot be labelled as such (and as a consequence, there is need for movement to break up the symmetrical construction). A moot point here is what exactly counts as non-accidental identity. We contend that members of a CC do qualify as non-accidentally identical since they are strictly required to be categorially and featurally fully identical (even though this identity is not conditioned by agreement).

<sup>11</sup> Hungarian has vowel harmony, thus the majority of words is either uniformly front-vowel or uniformly back-vowel.

#### 5.4 Morphology: in favour of post-syntactic lowering

We propose that the morphological properties of CCs (e.g., strict morphological parallelism) are a by-product of agreement of both members of a CC with a c-commanding head. The parallel morphology results from post-syntactic M(orphological)-Merger (Halle & Marantz 1993), whereby the suffixes are lowered to the heads post-syntactically. This explains (i) the requirement of strict morphological parallelism and (ii) possessor identity, as well as (iii) the pattern of dual affixation in Khanty. The mechanism of M-Merger is schematized in (35):



The post-syntactic character of affixation in CCs is supported by further evidence from Hungarian. In Hungarian, the 2SG subject agreement suffix that appears on verbs has two allomorphs (-sɛ vs. -l). Historically, the distribution of these allomorphs was phonologically conditioned, but as far as Modern Hungarian is concerned, the pattern is irregular: whether a stem selects -sɛ or -l is an idiosyncratic lexical property.

What happens if the two elements of a CC select different allomorphs of 2SG? Let us consider the case of, e.g., a CC *oszt-szoroz* (‘make calculations, consider one’s options’; lit: divide-multiply). To start with, if used in isolation, the 2SG forms for the two verbs are the following (Hungarian WebCorpus 2.0, Nemeskey 2020):

- |      |    |                              |              |               |      |            |           |
|------|----|------------------------------|--------------|---------------|------|------------|-----------|
| (36) | a. | <i>oszt-asɛ</i>              | divide-2SG   | standard form | 98%  | (386 hits) | Hungarian |
|      | b. | <sup>†/%</sup> <i>oszt-l</i> | divide-2SG   | dialectal     | 2%   | (8 hits)   |           |
| (37) | a. | <i>*szoroz-asɛ</i>           | multiply-2SG | unattested    | 0%   | (0 hits)   |           |
|      | b. | <i>soroz-l</i>               | multiply-2SG | standard form | 100% | (41 hits)  |           |

Based on these frequencies, on the assumption of independent probabilities, we can calculate the predicted frequencies of the possible CC forms:

- |      |    |                          |                 |           |
|------|----|--------------------------|-----------------|-----------|
| (38) | a. | <i>osztasɛ-szorozol</i>  | 98% (=98%*100%) | Hungarian |
|      | b. | <i>osztol-szorozol</i>   | 2% (=2%*100%)   |           |
|      | c. | <i>osztasɛ-szorozasɛ</i> | 0% (=98%*0%)    |           |
|      | d. | <i>osztol-szorozasɛ</i>  | 0% (=2%*0%)     |           |

The real, attested frequencies, however, are starkly different from the predicted ones (Hungarian WebCorpus 2.0, Nemeskey 2020):

- |      |    |                         |     |           |           |
|------|----|-------------------------|-----|-----------|-----------|
| (39) | a. | <i>osztasɛ-szorozol</i> | 56% | (15 hits) | Hungarian |
|      | b. | <i>osztol-szorozol</i>  | 33% | (9 hits)  |           |

- c. *osztas-<sup>z</sup>-osztas<sup>z</sup>*      11%      (3 hits)  
d. *osztol-<sup>z</sup>-osztas<sup>z</sup>*      0%      (0 hits)

The forms where the two suffixes are phonologically identical are vastly more frequent than one would expect assuming independent probabilities (33% vs. 2% and 11% vs. 0%). This distribution indicates that there is a strong preference for the suffixation to be identical not only featurally (2SG) but also in terms of phonological realization. This preference is strong enough to lead to the production of forms that are otherwise ungrammatical: \**osztas<sup>z</sup>* and <sup>†</sup>0% *oszt-ol* (the latter being ungrammatical for the large majority of speakers). Similar patterns are attested with *süts<sup>z</sup>-főzöl* (fry.2SG-cook.2SG ‘prepare a dish’), *húzol-balasztas<sup>z</sup>* (draw\_out.2SG-postpone.2SG ‘procrastinate’) and *ads<sup>z</sup>-ves<sup>z</sup>-el* (sell.2SG-buy.2SG ‘trade’). This preference for phonological identity strongly suggests that affixation in CCs takes place post-syntactically.

As the final piece of morphological evidence, let us consider a different type of morphological alternation. In Hungarian, many vowel-final nouns exhibit stem alternation conditioned by the suffix, as shown in (40), whereby the ACC- and PL-marked forms feature a closed-syllable form of the stem:

- (40) a. *kő*      stone (uncasemarked)      *fű*      grass (uncasemarked)  
*köv-et*      stone.ACC      *fűv-et*      grass.ACC  
*kő-nek*      stone.DAT      *fű-nek*      grass.DAT  
*köv-ek*      stone.PL      *fűv-ek*      grass.PL

Strikingly, if a vowel-final noun of this sort forms part of a CC, it seems to be immune to this alternation, as shown in (41):

- (41) a. *Mari fű-t*      - *fá-t*      *ígért.*  
Mary grass-ACC      tree-ACC      promised.  
b. \**Mari fűv-et*      - *fá-t*      *ígért.*  
Mary grass-ACC      tree-ACC      promised  
‘Mary promised everything imaginable.’ (lit. ‘Mary promised grass-tree.’)

While the precise analysis of this pattern is beyond the scope of this paper, it clearly lends further support to the idea of affixation being a late, post-syntactic process.<sup>12</sup> This, in turn, lends support to our analysis, which relies on M-Merger, and in particular, on the lowering of inflectional heads post-syntactically. A potential alternative analysis here would be to assume that the components of the CC already have their inflections on them at the point of Merge, with the (identical) feature bundles of the inflectional suffixes coalescing on the CC and undergoing checking/valuation *en bloc* against an outside probe. However, a very strict morphological parallelism between the components of the CC, including phonological identity (*modulo* vowel harmony) is more compatible with post-syntactic lowering.

<sup>12</sup> If one assumes that *-v-* in the ACC- and PL-marked forms actually belongs to the suffix (i.e., what we have is in fact suffix allomorphy and not stem alternation), then the pattern in (41) is simply another case of a strong preference for phonological identity in CC-suffixation. Within a stem-alternation approach, one needs to account for the fact that while the exponent of ACC (*-t*) appears on *fű* in (41a), ACC-triggered stem alternation fails to materialize. One may offer an ordering analysis: if lexical insertion precedes the copying of the suffix to the individual elements (at least in the case of nominal CCs), then the stem *fű* is actually not exposed to ACC, which explains that lack of stem alternation. Alternatively, one can analyze this in terms of a phonological uniformity requirement (or preference) driven by analogy: members of so-called echo words (to be discussed below) follow a strict phonological pattern – and this pattern might be extended by analogy to CCs too. But regardless of the exact analysis we pursue, the available evidence strongly suggests that the suffixation of CCs on both elements is a PF-phenomenon. (We would like to thank Péter Rebrus for an enlightening discussion of these matters.)

## 5.5 Meaning: superordinating & conjunctive CCs

We propose that CCs have a compositional semantics, with further differences between superordinating and conjunctive CCs. Namely, the calculation of the meaning of superordinating CCs involves a search for the immediate superordinate concept of the (intension of) the two elements, followed by existential or universal closure. In contrast, conjunctive CCs have a standard conjunctive interpretation.

Let us first consider superordinating CCs: here, the first step is a strictly local operation that searches for the common immediate superordinate concept of the two elements.<sup>13</sup> In hyponym-hypernym graph terms, this operation checks whether there is a node that immediately dominates the two elements; if so, at LF, the two-word sequence is replaced with the semantic feature bundle associated with that node, as is illustrated for Hungarian in (42), building on the context previously mentioned in (16) and (17):

(42) *süt-főz* ‘fry-cook’ [boil in fat/oil]-[boil in water] → [prepare dish]

The strict locality of the search explains why only synonyms, antonyms, taxonomic sisters, and reverses are allowed in superordinating CCs.<sup>14</sup>

Superordinating nominal CCs in Hungarian and Khanty obligatorily lack number marking (even when denoting a plurality). They are also strictly incompatible with a definite or indefinite article (in Hungarian). We argue that this indicates their status as bare nominals that lack the DP layer. This dovetails with their interpretation, which is quantificational rather than referential: superordinating CCs are obligatorily universal in Hungarian and universal or existential (depending on context) in Khanty.

Accordingly, we propose that, in Hungarian, (superordinating) nominal<sup>15</sup> CCs are Heimian indefinites that obligatorily undergo local universal closure (cf. (Postma 1995) and Corblin 2012):<sup>16</sup>

(43) *ügy-baj* -superordination->  $\lambda x.$ CONCERN( $x$ ) - $\forall$ -closure->  $\lambda Q \forall x.(\text{CONCERN}(x) \rightarrow Q(x))$   
 affair-trouble concern every concern

<sup>13</sup> The idea that some CCs (so-called collective CCs in the terminology of Wälchli (2005) denote a superordinate term of the elements is not in itself a novelty: this has been discussed by Wälchli (2005) and also by Szabolcsi (1976) in the context of Hungarian and Khanty. What is novel in our approach is (i) the extension of this idea to other types of CCs as well, (ii) the integration of this insight into a generative model of morphosyntax and semantics, (iii) the clarification of the two dichotomies (superordinating vs. conjunctive and regular vs. contextual), and (iv) the discussion of universal/existential closure.

<sup>14</sup> This mechanism also explains why superordinating CCs are typically made up of two elements (some examples of 3-element CCs do exist, such as *pénz-paripa-fegyver* money-horse-weapon ‘all necessary means’, *fű-fa-virág* grass-tree-flower ‘vegetation’ but these are very rare). Note that the syntactic mechanism of head-merge proposed above can in principle be iterated: in other words, nothing in syntax excludes the emergence 3-element, 4-element or n-element CCs. The reason why 3-element CCs are very rare and 3+-element CCs practically non-existent lies in the semantics of immediate superordination. Note that in an n-element CC, all the n elements have the same immediate superordinate term. From this, it follows that any two members of this n-member set also have the same immediate superordinate term. This means that going beyond two members is semantically superfluous and thus, uneconomical, resulting in a strong dispreference of 2+-member CCs.

<sup>15</sup> Note that in the case of non-nominal CCs, only superordination applies.

<sup>16</sup> This interpretational mechanism also explains the appearance of proper names in CCs (as long as the share an immediate superordinate term). Consider:

(i) *Mari bejárta Tolná-t-Baranyá-t.* Hungarian  
 Mary traversed Tolna-ACC-Baranya-ACC  
 ‘Mary traversed the whole country.’ (lit. ‘Mary traversed Tolna (county) and Baranya (county).’)

The derivation proceeds as follows:

(ii) *Tolna-Baranya* [county of Hungary named Tolna]-[county of Hungary named Baranya] -superordination-> [county of Hungary] -universal closure-> [every county of Hungary]

In superordinating CCs in Khanty, local universal closure is optional; in its absence, sentence-level existential closure ensues (Heim 1982):<sup>17</sup>

- (44) *kə:pə-fɛ:j* -superordination->  $\lambda x.$ DRINK( $x$ ) - $\exists$ -closure->  $\exists x.$ [... DRINK( $x$ )...]  
 coffee-tea drink some drink

In contrast with superordinating CCs, conjunctive CCs (found in Khanty), as we have seen above, receive a conjunctive interpretation (we repeat example (10) for convenience below):

- (45) *Mɛ: qut-əm-v βoqi-γən ɛ:mp-γən jv:βat-γən.* Khanty  
 1SG house-1SG-LAT fox-DU dog-DU come-PST.3DU  
 ‘A fox and a dog came to my house.’

The availability of conjunctive CCs in Khanty but not Hungarian seems to be correlated with two facts. First, syntactic coordination is a relatively recent development in Khanty, robustly attested only in mid-20th century (Borise & É. Kiss 2022), whereas it has been established in Hungarian since at least the 12th century (the date of the earliest extant written records of Hungarian). This means that in Khanty, until recently, in lieu of phrasal coordination of two nominals (e.g., *father and grandmother left*), one had to either resort to clausal juxtaposition (*father left, grandmother left*) or to co-compounding (*father-grandmother left*). The other crucial observation is that conjunctive CCs in Khanty contain number marking (DU or PL); whereas superordinating CCs (in both Khanty and Hungarian) lack number marking.<sup>18</sup>

Based on these observations, we hypothesize that, given that syntactic phrasal coordination is still relatively new in Khanty, CCs can still carry a vestigial function of coordination (*qua* conjunctive CCs), especially in informal/colloquial registers. This function, however, is being taken over by syntactic coordination. This process can be illustrated with the title of a well-known Khanty folk tale, which is translated as *A little bird and his older sister*. The title of the version recorded in 1901 is a CC, (46a), but the title of the one recorded in 1993 is an instance of coordination (46b):

- (46) a. *pi:tɨŋyali-γən o:pi-sɛ:-γən*  
 little.bird-DU older.sister-PROX-DU  
 ‘a little bird and his older sister’  
 b. *pi:təŋkəli pɛ:nə o:pi*  
 little.bird and older.sister  
 ‘a little bird and his older sister’

While no conjunctive CCs exist in Modern Hungarian, it is plausible to assume that Hungarian probably underwent the same process as Khanty, only earlier. In the absence of phrasal coordination, it is likely that the functional load of coordination was also in part carried by conjunctive CCs in Early

<sup>17</sup> It is unclear to us why Khanty allows this quantificational plasticity and why Hungarian does not. A natural assumption is that this is related to the mechanism of universal/existential quantification outside of CCs. However, since there is little formal work on quantification in Khanty, we have to leave this question to further research.

<sup>18</sup> At this point, one might wonder why phrasal coordination emerges in the first place, if its function can be covered by parataxis and conjunctive CCs. We would like to point out two considerations here. First, phrasal coordination is more economical than parataxis (*father and grandmother left the village* vs. *father left the village and grandmother left the village*). Secondly, co-compounding is a very limited strategy: the members need to be semantically closely related, need to have identical suffixes and cannot be modified separately (so *the car and the dog, your dog and my cat* or *one dog and three cats* are ineffable using CCs). Syndetic coordination is vastly more expressive, and, once a language has developed syndetic coordination, it is likely to edge out conjunctive co-compounding due to its strong competitive advantage.

Old Hungarian. However, the emergence and solidification of phrasal coordination has later led to the total extinction of conjunctive CCs.

### 5.6. Regular vs. contextual, superordinating vs. conjunctive

Finally, we would like to examine the exact relationship between the two CC dichotomies: regular vs. contextual, and superordinating vs. conjunctive. To recap, regular CCs are such that their elements are sisters in terms of semantic ontology, whereas with contextual CCs, a lower degree of semantic relatedness is sufficient — as long as the two elements form a pair in terms of the current discourse. Superordinating CCs are interpreted via superordination (replacement by the immediate shared superordinate concept) plus universal/existential closure; whereas conjunctive CCs receive a conjunctive interpretation. Furthermore, in Khanty, superordinating CCs lack number marking, whereas conjunctive CCs allow for it.

Hungarian lacks contextual CCs and it also lacks conjunctive CCs: thus, in Hungarian, all CCs are regular and superordinating. In Khanty, contextual CCs are necessarily conjunctive while regular CCs are open to both superordination and conjunction. Accordingly, combining these two dichotomies into a 2x2 matrix produces the following picture:

|            | superordinating | conjunctive |
|------------|-----------------|-------------|
| regular    | ✓               | ✓           |
| contextual | ✗               | ✓           |

Table 1

This overall picture makes sense in terms of our current proposal. Superordination requires that the two elements be ontological sisters (otherwise, the search for an immediate common superordinate term fails): this means that only regular CCs can undergo superordination. The conjunctive interpretation makes no such requirement: hence, contextual CCs are compatible with a conjunctive interpretation.

Finally, it is worth pointing out that, in Khanty, a pair of ontological sisters is amenable to both types of compounding (superordinating and conjunctive). Whether or not such CCs are to be interpreted superordinatingly or conjunctively can be read off from the presence or absence of dual marking. Specifically, a dual-marked co-compound can only be interpreted conjunctively (even if the two elements happen to share an immediate superordinate concept):

- (47) *i:mi-yən*                      *i:ki-yən*                                              Khanty  
 older.woman-DU – older.man-DU ...  
 ✓ i. ‘an old(er) woman and an old(er) man...’  
 ✗ ii. ‘all/some old(er) people ...’

Conversely, the lack of number marking enforces a superordinating interpretation:

- (48) *xɐ:nti*    *xo*                      *xɐ:nti*    *ne:*                                              Khanty  
 Khanty man    Khanty woman  
 ✗ i. ‘a Khanty man and a Khanty woman’  
 ✓ ii. ‘(all/some) Khanty people’

We tentatively hypothesize that, during the syntactic derivation, the presence of number marking might interfere with the search for a superordinate concept, which leads to the CCs acquiring a meaning equivalent to that of coordination of two items (conjunctive CC). This dovetails with the evidence from Hungarian, where syntactic coordination has been available for a longer time, CCs do not allow for conjunctive readings and do not have number marking. We leave in-depth treatment of the



correlation between the availability of number marking and the interpretation of a CC for further research.

## 6. What CCs are not

### 6.1: CC(V)=/SVC: Verbal CCs are not serial verb constructions

Before concluding the paper, we would like to discuss how CCs (which are two-headed phrases in our analysis) are fundamentally different from another, better-known phenomenon that has also been analysed as involving a two-headed phrase: so called serial verb constructions (SVCs).

Serial verb constructions (or at least one subtype of them, cf. Aikhenvald & Dixon 2005 and Lovestrand 2021 for recent overviews) have been analysed as two-headed verb phrases where the two verbs share an object (Baker 1989):<sup>19</sup>

- (49) *Ikán jẹ ìwé Adio run*  
 termite eat book Adio destroy  
 ‘Termites ate and thus destroyed Adio’s book’ (Yoruba, Ekundayo & Akinnaso 1983)<sup>20</sup>

However, serial verb constructions are very different from CCs both in terms of their syntax and their semantics. Considering syntax first, one trivial but important difference is that co-compounding is a cross-categorical phenomenon (N-N, V-V, Adj-Adj, Adv-Adv, P-P etc.) while serial verb constructions are limited to verbs. Secondly, as we have seen, in CCs, the two verbs are strictly adjacent, whereas in SVCs, the canonical position of the object is between the two verbs. Thirdly, in verbal CCs, the two verbs need to have identical theta-grids; but not so in SVCs, where it is possible to combine a transitive verb (agent+theme) with a three-argument verb (agent+theme+goal/location):

- (50) *Bàbá mi ra ẹwù bìn mi.*  
 father my buy garment present(verb) me  
 ‘My father brought me a garment.’ (Yoruba, Oyelaran 1982)

In terms of semantics, we have seen that in a superordinating CC, the two elements share an immediate superordinate term, and in conjunctive CCs, they need to be reasonably closely related semantically. In contrast, the elements of SVCs do not need to share an immediate superordinate term. In so-called symmetric SVCs, there is no semantic restriction whatsoever, and “the semantic interpretation is typically [that] of two closely associated sequential activities, sometimes with a cause–effect interpretation” (Lovestrand 2021: 120):

- (51) *Wọ̀n bú omi mu.*  
 they pour water drink(trans)  
 ‘They poured water and drank it.’ (Yoruba, Carstens 1988a)

In so-called asymmetric SVCs, one of the verb slots is restricted to a particular verb class (e.g. directional verbs denoting path of motion):

- (52) *Titi rìn lẹ̀ nì, kò sáré lẹ̀.*  
 Titi walk go is, no run go  
 ‘Titi left walking, not running.’ (Yoruba, Ekundayo & Akinnaso 1983)

<sup>19</sup> For alternative approaches to SVCs (which do not involve two-headed phrases) cf. Collins (1997) and Aboh (2009), a.o.

<sup>20</sup> All examples in this section are taken from Baker (1989), with the original sources provided in brackets.

Given these substantial empirical differences, it should come as no surprise that these two constructions — SVCs and CCs — have been modelled differently. In Baker (1989), SVCs are analyzed as doubly-headed VPs: however, crucially, the two Vs stay distinct, as opposed to in CCs, where (in our analysis) the two Vs are head-merged and form a complex head:

- (53)      a. SVCs (Baker 1989)                      b. CCs (current proposal)
- ```

graph TD
    VP --> V_prime[V']
    V_prime --> V1[V1]
    V_prime --> OBJ[OBJ]
    V_prime --> V_prime2[V']
    V_prime2 --> V2[V2]

```

```

graph TD
    VP --> V_prime[V']
    V_prime --> V[V]
    V_prime --> OBJ21[OBJ21]
    V --> V1[V1]
    V --> V2[V2]

```

Furthermore, in SVCs, there are two distinct heads which share an object but are otherwise independent: they stay syntactically, semantically and indeed phonologically distinct all along the derivation. In CCs, the heads are syntactically merged early on and act as complex head as far as syntax and morphology are concerned; also, in terms of their semantics, their individual denotations may not always be preserved: in superordinating CCs, they are replaced by a superordinate term.

6.2. CCs vs. coordinative bare singular nominals in Dutch

Postma (1995) discusses an interesting pattern in Dutch which bears some similarity to CCs:

- (54) *Zij hebben zich met hand en tand verzet.*
they have themselves with hand and tooth resisted
‘They resisted with everything (they had)’ (lit. ‘they resisted with hand and tooth’)
- (55) *Het schip verging met man en muis.*
the ship went.down with man and mouse
‘The ship sank with everyone (every living thing) on it.’

The examples discussed are the following:

- (56)
- | | | |
|---------------------------|------------------|--|
| a. <i>kind noch kraai</i> | child nor crow | ‘no company at all’ |
| b. <i>man en muis</i> | man and mouse | ‘everyone’ |
| c. <i>hand en tand</i> | hand and tooth | ‘every physical means at one’s disposal’ |
| d. <i>buis en haard</i> | house and hearth | ‘all one’s possessions’ |
| e. <i>men en macht</i> | man and power | ‘every means at one’s disposal’ |
| f. <i>buid en haar</i> | skin and hair | ‘all parts’ |
| g. <i>steen en been</i> | stone and bone | ‘all one’s strength’ |

Similarly to (nominal) CCs in Hungarian, the elements must be bare singular. The requirement of semantic relatedness is more relaxed (‘child’ and ‘crow’ and, especially, ‘man’ and ‘power’ are not very close in semantic ontology), while the requirement of phonological similarity is more pronounced (apparently, the elements must rhyme either in the onset or the coda). Possibly because of this strict phonological requirement, the process is much less productive than co-compounding, being limited to

²¹ In this schematic rendering, we show the object as the complement to the right of V for simplicity (as opposed to, e.g., the specifier of VP; or complement to the left in a head-final language), but nothing in our proposal hinges on this choice.

a handful of fixed expressions. Note also that in contrast to CCs in Hungarian and Khanty, prepositions (which come closest to suffixes/postpositions in Hungarian and Khanty) do not get duplicated: *met hand and (*met) tand*.

In terms of semantics, even though there is an overt coordinator between the two elements, the interpretation at LF is not conjunctive. Postma characterizes this phenomenon as ‘universal quantification under dummy coordination’.

As far as we could surmise, Postma (1995) does not address the question of superordination (i.e., how *man en muis* receives the interpretation of ‘person/living soul’). To explain the universal quantificational interpretation, Postma (1995) develops a theory which is a unique blend of Saussurean semiotics and X-bar-theory, assuming that every nominal expression has a separate syntactic slot for both the lexical element (N^0 , equated by Postma with the *signifié*) and its ‘meaning’ (the *signifiant*), the latter slot being filled by a PRO element: $[\text{PRO } N^0]_{NP}$. Further, Postma argues that the interpretation of a nominal (denotational vs. quantificational) depends on how high this PRO element has moved within the DP. In the constructions under examination, the bare singular nature of the nominals excludes a denotational interpretation and forces movement, which leads to a quantificational interpretation. In terms of why such an interpretation is only available to (dummy) coordinate structures, Postma argues that all universally quantified structures have an underlyingly coordinating syntactic structure (i.e. *alle huizen* ‘all houses’ is underlying *huizen en al*).

In our view, Postma’s proposal relies on assumptions which are overly broad and ad-hoc at the same time. In addition to this, it also fails to provide any motivation of the semantic and phonological constraints that these constructions are subject to. While it is firmly not our intention to provide, in this paper, an analysis of coordinative bare singular nominals in Dutch, it seems plausible to argue that, while syntactically different, they are rather similar to superordinating CCs in terms of their semantics. Superordination is facilitated by semantic relatedness and phonological similarity, whereas quantificational closure obtains because of the impossibility of referential interpretation (due to the lack of a DP layer, indicated by the absence of both a determiner and plural marking). Notwithstanding this similarity in interpretation, we believe that CCs and coordinative bare singular nominals are different phenomena, for the following reasons, which set CCs apart from coordinative bare singular nominals: (i) CCs are not limited to nominals, (ii) CCs never contain an overt conjunction, (iii) CCs can have a conjunctive interpretation, and (iv) co-compounding is productive whereas coordinative bare singular nominals are idiomatic.

6.3 CCs vs. Lexical coordination

Let us consider another phenomenon that might seem similar to co-compounding: lexical (i.e., non-phrasal) coordination — that is, the coordination (by way of an overt coordinator) of lexical heads (X^0 elements), as illustrated below:

- (57) *Le président* [[*apprécie_V* et *approuve_V*] *votre proposition*] *VP*.
the president appreciates and approves your proposal
‘The president appreciates and approves your proposal.’

- (58) *Indiquez* [*votre/vos* [*nom_N* et *prénom_N*]] *DP*.
indicate your_{SG}/your_{PL} surname and first.name
‘Provide your surname and first name.’

While some authors have argued that putative cases of lexical coordination should rather be analyzed as phrasal coordination plus ellipsis (Kayne 1994, Beavers and Sag 2004, a.o.), the status of lexical coordination as a separate phenomenon has been defended vigorously by others (Abeillé 2006, Abeillé & Chaves, 2021 a.o.). In this paper, we will stay agnostic as to the proper analysis of lexical coordination. Rather, we will focus on the considerable empirical differences between CCs and lexical coordination, and will argue that co-compounding and lexical coordination are clearly distinct phenomena.

In terms of syntax, CCs and lexical coordination are similar in that the two elements share their complements and have identical theta grids. Beyond this, considerable differences obtain. Lexical coordination always²² features an overt coordinator, while CCs never do. Also, a pair of lexically coordinated nominals are happy to have a shared determiner whereas nominal CCs clearly lack a DP layer.

In terms of semantics, lexical coordination yields a referential expression, whereas superordinating CCs are non-referential. The interpretation is also different: lexical coordination always²³ receives a conjunctive interpretation, whereas the interpretation of a superordinating CC involves superordination and universal/existential closure. The degree of semantic relatedness required between the elements is also much lower when it comes to lexical coordination.

To conclude, we believe that what has been discussed under the rubric of lexical coordination is indeed coordination (of heads or phrases, depending on one's theoretical persuasion): there is an overt coordinator, the interpretation is always conjunctive, the semantic relatedness is looser, the elements retain more of their independence throughout the derivation. As opposed to this, as we have argued above, co-compounding is an instance of head-merge (i.e., not coordination).

7. Further research directions

An interesting question is whether the analysis proposed in this paper can be extended to so-called 'echo words' in Hungarian. Their syntactic behaviour is identical to CCs in terms of the two elements being obligatory adjacent, lacking an overt coordinator, sharing complements, having identical suffixation etc. However, while in CCs, both elements are independent lexemes in their own right (albeit one of the two might be archaic), in echo words, the first element is a non-word (a slightly modified version of the second element in terms of its phonological makeup, cf. Sóskuthy (2012), Patay (2018)). In terms of their meaning, echo words are interpreted as diminutives of the second element.

- (59) a. *riheg-röhög*
nonword-laugh
'gigglev'
b. *ici-pici*
nonword-small

²² Anne Abeillé (p.c.) provides *allez-retour* (a ticket valid in both directions) as an apparent counterexample. However, this seems to be an isolated phenomenon: all other instances of low coordination that we could think of seem to be completely unacceptable without a coordinator: **Le président apprécie-approuve votre proposition.*, **Indiquez votre nom-prénom.*, **mes soeurs-frères* 'my sisters and brothers' etc.

²³ Anne Abeillé (p.c.) points out that some instances of low coordination are compatible with a superordinating reading, such as *mes mère et père* 'my parents' (lit. 'my mother and father'). However, all these instances happen to be such that superordination and conjunction actually lead to the same interpretation: 'my parents' = 'my mother and my father' (under the prototypical scenario). And note also that in the majority of instances, low coordination does not involve superordination (the elements do not even share an immediate superordinate term, and the reading is unequivocally conjunctive). Thus, on balance, we believe it is unwarranted to assume that superordination happens in the case of low coordination.

‘tiny’

It seems to us that echo words can probably be analyzed as a subtype of co-compounding. E.g., assuming that the corrupted forms have the same interpretation as the originals, *riheg-röbhög* would trivially receive the interpretation of [laugh] (as the closest superordinate term of [laugh] and [laugh] is [laugh]). However, this would leave unexplained the diminutive interpretation. One might offer an analysis in terms of ordering: the first step being co-compounding producing *röbhög-röbhög* ‘laugh’, and the second step being diminution (which for some reason is phonologically manifested on the first element only), producing *riheg-röbhög* ‘giggle’. This, however, begs the question of why we do not simply produce **riheg* ‘giggle’ (a form that is actually non-existent) without taking the extra step of co-compounding. We have to leave a more thorough analysis of these questions to future work.

8. Conclusion

Based on evidence from Khanty and Hungarian, we conclude that the term ‘coordinative compounds’ is something of a misnomer, as CCs, in fact, lack coordination and they are not compound words stored as such in the lexicon. Instead, we argued that co-compounding is a syntactic operation involving the Merge of two syntactic heads which share functional projections. We have shown that co-compounding (the existence of two-headed phrases) is compatible with standard generative assumptions, as long as the two heads meet certain requirements (identical category and thematic and selectional requirements, close semantic relationship): exactly the ones that CCs are, in fact, subject to. Further, we argued that the matching morphology on the elements of a CC results from post-syntactic lowering and that the meaning of superordinating CCs is derived via a search for an immediate superordinate concept at LF and universal/existential closure. We have argued that conjunctive CCs (available only in Khanty), the meaning of which is equivalent to that of coordination, are a vestige from old Khanty that lacked syntactic coordination. They are being replaced by syntactic coordination but are also likely protected from quick extinction by their distinct morphosyntactic properties (number marking).

Acknowledgements

We are grateful to the speakers of Surgut Khanty who provided the data for their time and patience. For helpful comments and advice at various stages of this project, we would like to thank Márta Csepregi, Katalin É. Kiss, Marcel den Dikken, and other colleagues at the Hungarian Research Centre for Linguistics; Anne Abeillé, Olivier Bonami, and other colleagues at LLF; the audience at the Uralic Information Centre talk, and the audiences and reviewers at ICHS 16, LeiBieCos 1, SOUL 5, and Form and Meaning of Coordination. All remaining errors are ours.

Funding information

This research has been supported by Grants 129921 and 135958 of the National Research, Development and Innovation Fund of Hungary, the Bolyai scholarship of the Hungarian Academy of Sciences, the Bolyai Plus scholarship of the New National Excellence Programme of the Ministry of Innovation and Technology of Hungary, and funding from the European Union’s Horizon Europe research and innovation programme under the Marie Skłodowska-Curie grant agreements No 101109402 and No 101199275

References

- Aboh, Enoch Oladé. 2009. Clause Structure and Verb Series. *Linguistic Inquiry* 40(1). 1–33. <https://doi.org/10.1162/ling.2009.40.1.1>.
- Aikhenvald, Alexandra Y. & R. M. W. Dixon. 2005. *Serial Verb Constructions: A Cross-Linguistic Typology*. OUP Oxford.
- Baker, Mark C. 1989. Object Sharing and Projection in Serial Verb Constructions. *Linguistic Inquiry*. The MIT Press 20(4). 513–553.
- Bauer, Laurie. 2023. Coordinative Compounding, Including Dvandva. In Peter Ackema, Sabrina Bendjaballah, Eulàlia Bonet & Antonio Fábregas (eds.), *The Wiley Blackwell Companion to Morphology*, 1–31. John Wiley & Sons.
- Borise, Lena & Katalin É. Kiss. 2022. The emergence of conjunctions and phrasal coordination in Khanty. *Journal of Historical Linguistics* Online First. <https://doi.org/10.1075/jhl.21016.kis>.
- Carstens, Vicki. 1988. Yoruba serial verbs. In *2nd Niger-Congo Syntax and Semantics Workshop, MIT, Cambridge, Massachusetts*.
- Chomsky, Noam. 1970. Remarks on Nominalization. In Roderick A. Jacobs & Peter S. Rosenbaum (eds.), *Readings in English Transformational Grammar*, 184–221. Waltham, MA; Toronto; London: Ginn and Company.
- Chomsky, Noam. 1993. *Lectures on government and binding*. Walter de Gruyter.
- Chomsky, Noam. 2013. Problems of projection. *Lingua* 130. 33–49.
- Collins, Chris. 1997. Argument Sharing in Serial Verb Constructions. *Linguistic Inquiry*. The MIT Press 28(3). 461–497.
- Den Dikken, Marcel. 2004. Agreement and “clause union.” In Katalin É. Kiss & Henk van Riemsdijk (eds.), *Verb Clusters: A study of Hungarian, German and Dutch*. John Benjamins Publishing.
- Di Sciullo, Anna Maria. 2002. *Asymmetry in Grammar*. Amsterdam ; Philadelphia: John Benjamins Pub.
- Di Sciullo, Anna Maria. 2005. *Asymmetry in Morphology*. Cambridge, MA: MIT Press.
- É. Kiss, Katalin. 2002. *The Syntax of Hungarian*. Cambridge: Cambridge University Press.
- Ekundayo, S. Ayotunde & F. Niyi Akinnaso. 1983. Yoruba serial verb string commutability constraints. *Lingua* 60(2). 115–133. [https://doi.org/10.1016/0024-3841\(83\)90070-0](https://doi.org/10.1016/0024-3841(83)90070-0).
- Munn, Alan Boag. 1993. *Topics in the syntax and semantics of coordinate structures*. University of Maryland at College Park PhD Thesis.
- Oravecz, Csaba, Tamás Váradi & Bálint Sass. 2014. The Hungarian Gigaword Corpus. In *Proceedings of the 9th International Conference on Language Resources and Evaluation (LREC'14), Reykjavik, Iceland*, 1719–1723. European Language Resources Association (ELRA).
- Oyelaran, Olasope O. 1982. On the scope of the serial verb construction in Yoruba. 13(2). 109–146.
- Patay, Fanni. 2018. A magyar ikerszók optimalitáselméleti elemzése. In György Scheibl (ed.), *LingDok 17. Nyelvészdoktoranduszok dolgozatai*, 67–84. Szeged: Szegedi Tudományegyetem Nyelvtudományi Doktori Iskola.
- Piñón, Christopher. 1995. Around the progressive in Hungarian. *Approaches to Hungarian* 5. 153–190.
- Postma, Gertjan. 1995. Zero Semantics — The syntactic encoding of quantificational meaning. *Linguistics in the Netherlands* 12. 175–190. <https://doi.org/10.1075/avt.12.17pos>.
- Scalise, Sergio, Antonio Fábregas & Francesca Forza. 2009. Exocentricity in compounding. *Gengo Kenkyu (Journal of the Linguistic Society of Japan)* 135. 49–84.
- Sóskuthy, Márton. 2012. Morphology in the extreme: echo-words in Hungarian. In Ferenc Kiefer & Zoltán Bánréti (eds.), *Twenty Years of Theoretical Linguistics*, 123–143. Budapest: Tinta Kiadó.
- The Ob-Ugric Database (OUDB). The Ob-Ugric Database (OUDB). <http://www.babel.gwi.uni-muenchen.de/index.php?navi=linguistic&abfrage=linguistic>. (17 September, 2024).
- Wälchli, Bernhard. 2005. *Co-Compounds and Natural Coordination*. Oxford: Oxford University Press.